

## Algo Trading Game

Throughout the week, you guys will be designing algos to trade against each other and with bots on a synthetic exchange. The focus of the first day is primarily for you guys to get connected to the exchange, send orders, and create an algorithm. The underlying through the week will generally remain the same: this will settle to the sum of  $N$  numbers revealed after some time. There will be more complexities added as the week progresses.

For now, the initial underlying works as follows:

- Let  $a \sim \text{floor}[\text{LogNormal}(1, 0.8)]$  and  $w \sim 1 + \text{floor}[\text{LogNormal}(0.5, 0.7)]$
- Let  $b = a + w$
- We draw random variable  $X_i \sim \text{Uniform}(a, b)$   $N$  times over some time period  $T$
- The underlying settles to the sum of  $X_i$

The first day, each game will run for 5 minutes, with numbers revealed every 30 seconds. Note: once you start the game, it will not stop until the game ends.

Some other notes:

- A sample notebook "getting\_started.ipynb" made by Claude has been provided. This will hopefully reduce the time to get started. Hint: not all exchange messages provided by the exchange are given in the notebook
- The parameters of the game are not guaranteed to stay the same, though the distributions will
- There are bots that will trade with each other, and with you
- Additional markets (which may include derivatives) will be added as the week goes on
- Everyone (including the bots), has a position limit of  $\pm 100$ , which is subject to change
- The goal of this is to create an algorithm that can consistently make money

I will be absent Day 1. If you have questions, try to figure it out yourself, otherwise you can send a message in Slack.